

Supplementary Data

Metabolic Reprogramming in the *opaque2* Maize Endosperm:
A Genome-Scale Model Reveals Coordinated Flux Rewiring
for Lysine Overaccumulation

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Supplementary Tables S1–S22 contain all quantitative tabular results supporting the main manuscript, organized by Results section.

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1 Results 1.1: Wild-type and *o2* Biomass Flux

Table S1. Comparison of maximum biomass flux between wild-type (WT) and *o2* mutant endosperm across development.

DAP	WT Biomass	O2 Biomass	Difference	Fold Change	% Change
6	5.99×10^{-4}	5.87×10^{-4}	-1.27×10^{-5}	0.979	-2.1
8	5.66×10^{-4}	6.62×10^{-4}	$+9.63 \times 10^{-5}$	1.170	+17.0
10	6.28×10^{-4}	6.39×10^{-4}	$+1.06 \times 10^{-5}$	1.017	+1.7
12	7.27×10^{-4}	6.71×10^{-4}	-5.58×10^{-5}	0.923	-7.7
15	1.60×10^{-3}	1.33×10^{-3}	-2.75×10^{-4}	0.828	-17.2
18	1.23×10^{-3}	1.61×10^{-3}	$+3.79 \times 10^{-4}$	1.307	+30.7
22	3.60×10^{-4}	3.56×10^{-4}	-4.80×10^{-6}	0.987	-1.3
30	1.75×10^{-4}	5.97×10^{-5}	-1.16×10^{-4}	0.340	-66.0

2 Results 1.2: Lysine Biosynthetic Pathway Flux

Table S2. Flux through DAP decarboxylase (R00451, final step of lysine biosynthesis) across development.

DAP	WT Flux	O2 Flux	Difference	Fold Change
6	2.51×10^{-6}	3.73×10^{-6}	$+1.22 \times 10^{-6}$	1.49
8	2.37×10^{-6}	4.05×10^{-6}	$+1.68 \times 10^{-6}$	1.71
10	2.63×10^{-6}	2.68×10^{-6}	$+5.0 \times 10^{-8}$	1.02
12	3.05×10^{-6}	2.81×10^{-6}	-2.4×10^{-7}	0.92
15	6.73×10^{-6}	5.80×10^{-3}	$+5.80 \times 10^{-3}$	862
18	5.18×10^{-6}	6.77×10^{-6}	$+1.59 \times 10^{-6}$	1.31
22	1.51×10^{-6}	1.251	+1.251	$> 8 \times 10^5$
30	~ 0	~ 0	~ 0	—

3 Results 1.3: Lysine Catabolism Bounds

Table S3. Upper bounds (v_{max}) for lysine degradation reactions, showing reduced catabolic capacity in the *o2* mutant.

DAP	R00716 (LKR alt)			R03102 (AAS DH)		
	WT v_{max}	O2 v_{max}	% Diff	WT v_{max}	O2 v_{max}	% Diff
6	2.951	2.809	−4.8	0.189	0.157	−16.9
8	2.545	2.612	+2.6	0.188	0.152	−19.1
10	2.585	2.439	−5.6	0.164	0.149	−8.7
12	2.401	2.399	−0.1	0.195	0.191	−2.0
15	3.688	2.374	−35.6	0.323	0.238	−26.2
18	3.066	2.433	−20.6	0.278	0.226	−19.0
22	6.357	2.987	−53.0	0.334	0.228	−31.8
30	5.520	3.929	−28.8	0.278	0.225	−18.8

4 Results 1.4: Lysine Flux Balance

Table S4. Lysine flux balance: total production, consumption, and turnover ratio (O2/WT).

DAP	WT Production	O2 Production	O2/WT Ratio	Net Balance
6	7.53×10^{-6}	9.92×10^{-6}	1.32	≈ 0
8	7.11×10^{-6}	1.09×10^{-5}	1.53	≈ 0
10	7.89×10^{-6}	8.04×10^{-6}	1.02	≈ 0
12	9.15×10^{-6}	8.43×10^{-6}	0.92	≈ 0
15	2.02×10^{-5}	1.74×10^{-2}	862	≈ 0
18	1.55×10^{-5}	2.03×10^{-5}	1.31	≈ 0
22	4.53×10^{-6}	3.752	$> 8 \times 10^5$	≈ 0
30	7.4×10^{-7}	2.5×10^{-7}	0.34	≈ 0

5 Results 1.5: Lysine Exchange Flux

Table S5. Lysine exchange reaction flux (Exchange_C00047[K,L]) across development.

DAP	WT Exchange	O2 Exchange	Difference
6	0	0	0
8	0	0	0
10	0	0	0
12	0	0	0
15	0	5.80×10^{-3}	$+5.80 \times 10^{-3}$
18	0	0	0
22	0	1.251	+1.251
30	-7.4×10^{-7}	-2.5×10^{-7}	$+4.9 \times 10^{-7}$

6 Results 1.6: Pathway-Level Flux Comparison

Table S6. Pathway-level flux comparison (total absolute flux, percent change O2 vs. WT) at key developmental stages.

Pathway	DAP 6	DAP 15	DAP 22	DAP 30
Lysine Biosynthesis	+72.8%	+22.8%	+18,222%	+195,928%
Lysine Degradation	—	—	—	—
Aspartate Pathway	+72.8%	+5.7%	+9,280%	+195,928%
TCA Cycle	+14.5%	+72.4%	+2,163%	−66.2%
Glycolysis	+472.3%	−41.9%	+196.7%	−96.0%
Amino Acid Transport	+1.9%	+20,289%	$+1.95 \times 10^7\%$	−65.9%
Starch Metabolism	−2.1%	−17.2%	−1.3%	−65.9%
Pentose Phosphate	−2.1%	−76.5%	+18.5%	−80.5%

7 Results 1.7: Amino Acid Biomass Coefficients

Table S7. Biomass stoichiometric coefficients for aspartate-family amino acids (identical in WT and O2 at all DAPs).

Amino Acid	Biomass Coefficient (mmol/unit)
Aspartate (C00049)	−0.045010
Threonine (C00188)	−0.008145
Lysine (C00047)	−0.004194
Methionine (C00073)	−0.001789
Isoleucine (C00407)	−0.000933

8 Results 1.8: Rewiring Index and Carbon Allocation

Table S8. Rewiring Index ($\text{RI} = \sum_j |v_j^{\text{O2}} - v_j^{\text{WT}}|$) and carbon allocation to lysine across development.

DAP	Rewiring Index	WT % C to Lys	O2 % C to Lys	O2/WT Ratio
6	70.5	< 0.01	0.00	—
8	17.9	< 0.01	0.00	—
10	17.2	< 0.01	< 0.01	—
12	3.1	< 0.01	< 0.01	—
15	61.2	< 0.01	0.04	$\gg 1$
18	151.5	< 0.01	< 0.01	—
22	228.1	< 0.01	4.31	$\gg 100$
30	39.8	0.00	0.00	—

9 Results 1.9: Differential Reaction Analysis

Table S9. Number of significantly different reactions at each developmental stage.

DAP	Reactions with $ \Delta v > 10^{-6}$
6	373
8	455
10	331
12	367
15	489
18	468
22	314
30	443

Table S10. Top 10 most differentially active reactions between WT and *o2* mutant.

Reaction	Max $ \Delta v $	Avg $ \Delta v $
R00307[K,c]	27.16	6.80
Exchange_C00031[K,L]	25.44	6.92
ExB_C00031[K]	25.44	6.92
PhloemImport_C00095[K,c]	23.95	7.09
PhloemTransport_C00095[K,c]	23.95	7.09
Exe4[K]	11.04	1.47
Trans4[K]	11.04	1.47
cpTransport_C00011[K]	10.79	1.50
ExB_C00089[K]	7.09	0.98
R00801[K,c]	7.09	0.98

Table S11. Number of uniquely active reactions in each genotype across development.

DAP	WT Active	O2 Active	Unique WT	Unique O2
6	550	617	41	108
8	600	622	67	89
10	584	564	60	40
12	571	560	49	38
15	585	571	90	76
18	570	606	67	103
22	538	553	82	97
30	518	512	103	97

10 Results 1.10: Carbon and Nitrogen Budget Analysis

Table S12. Carbon budget comparison at DAP 22: aggregate flux sums for central carbon pathway metabolites (cytosol + plastid).

Pathway Category	WT Flux Sum	O2 Flux Sum	Fold Change
Glucose import	25.96	31.24	1.2×
Glycolysis	11.59	41.24	3.6×
TCA cycle	1.09	29.80	27.4×
Pyruvate node	2.24	19.02	8.5×

Table S13. Nitrogen budget comparison at DAP 22: aggregate flux sums for nitrogen-containing pathway metabolites (cytosol + plastid).

Pathway Category	WT Flux Sum	O2 Flux Sum	Fold Change
Nitrogen donors (Glu + Gln)	0.14	8.03	57×
Aspartate family	0.07	7.81	111×
Lysine pathway	0.07	11.56	165×
Lysine degradation	0.00	2.50	—
BCAA	1000.07	1001.47	1.0×
Aromatic AA	0.00	2.85	—
Other AA	1000.00	1000.00	1.0×

Table S14. Key trade-offs at DAP 22: metabolites with the largest flux sum decreases in the *o2* mutant, representing resources redirected away from their normal wild-type pathways.

Metabolite	WT Φ	O2 Φ	$\Delta\Phi$
D-Fructose [boundary]	24.180	16.463	-7.717
Glucose-6-P (C00092) [cyt]	1.291	0.000	-1.291
PPi [cyt]	1.208	0.010	-1.198
AMP [cyt]	1.203	0.010	-1.193
D-Fructose [cyt]	24.481	23.850	-0.631
DHAP [pla]	0.135	0.005	-0.130
GAP [pla]	0.130	0.005	-0.125
1,3-bisP-glycerate [pla]	0.130	0.005	-0.125
Palmitoyl-CoA [pla]	0.010	0.000	-0.010
Palmitate [pla]	0.010	0.000	-0.010

11 Results 1.11: Flux Bound Analysis

Table S15. Top reactions where flux bound differences between WT and O2 at DAP 22 result in the largest actual flux changes, revealing the regulatory drivers of metabolic reprogramming.

Reaction	WT v_{max}	O2 v_{max}	WT Flux	O2 Flux
R00801 (Malate DH) [cyt]	10.86	15.42	0.301	7.388
R00342 (Asp aminotransferase) [cyt]	7.41	10.18	0.120	-4.204
R01512 (Enolase) [cyt]	17.24	20.36	-0.102	-4.367
R00216 (Malate DH) [pla]	4.16	4.33	0.197	4.326
R00480 (Aspartate kinase) [pla]	6.58	5.70	0.000	1.251
R01213 (DAP aminotransferase) [pla]	37.79	62.45	0.070	1.470
R02611 (Phe biosynthesis) [pla]	3.71	4.11	0.000	1.424

12 Results 1.12: Engineering Target Identification

Table S16. Top knockout targets for lysine enhancement at 18 DAP (restricted to reactions with GPR associations). All targets maintain 100% wild-type biomass.

Rank	Reaction	Function	R00451 FC	Gene(s)
1	R00216[K,p]	Pyruvate DH (plastid)	16,234×	5 isozymes
2	R00282[K,c]	Acetyl-CoA carboxylase	28×	1 gene

Table S17. Representative overexpression targets for lysine enhancement at 18 DAP ($2 \times v_{max}$). Only reactions with GPR associations are shown.

Reaction	Function	R00451 FC	Biomass Change	Gene(s)
R01280[K,p]	Chorismate synthesis	$1.096 \times$	+0.006 ppm	13 isozymes
R00883[K,c]	Phosphofructokinase	$1.096 \times$	+0.006 ppm	2 isozymes
R00512[K,c]	Pyruvate kinase (cytosol)	$1.096 \times$	+0.006 ppm	4 isozymes
R01015[K,c]	Glycolysis (GAPDH)	$1.096 \times$	+0.006 ppm	15 isozymes
R00667[K,m]	TCA cycle (mitochondrial)	$1.096 \times$	+0.006 ppm	2 isozymes

Table S18. o2-bound mimicry targets at 18 DAP.

Reaction	Function	FC	v_{max} Change
MR00422[K,p]	Lipid metabolism (plastid)	$1.096\times$	-14.2%
MR00732[K,m]	Mitochondrial metabolism	$1.096\times$	+26.9%
R00342[K,p]	Transketolase (plastid)	$1.096\times$	+102.1%

Table S19. Top combination strategies for lysine enhancement (18 DAP), restricted to reactions with GPR associations. All achieve 100% biomass retention.

Combination Strategy	R00451 FC	Biomass
MIM:MR00422 + KO:R00216 (Lipid ↓ + Pyr DH block)	16,234×	100%
OE:R01280 + KO:R00216 (Chorismate ↑ + Pyr DH block)	16,234×	100%
OE:R00883 + KO:R00216 (PFK ↑ + Pyr DH block)	16,234×	100%
MIM:R00342 + KO:R00282 (TKT ↑ + ACC block)	28×	100%
OE:R01280 + KO:R00282 (Chorismate ↑ + ACC block)	28×	100%
KO:R00216 + KO:R00282 (Pyr DH block + ACC block)	16,234×	100%

Table S20. Gene-level engineering targets for lysine enhancement in wild-type maize endosperm. Genes were identified from GPR rules in the endosperm genome-scale model. Intervention type: KO = knockout; OE = overexpression; MIM = *o*2-bound mimicry. All listed reactions have verified GPR associations; reactions without gene associations (e.g., inter-compartment transport steps) are excluded.

Reaction	Function	Type	FC	#G	Gene IDs
Tier 1: High-Impact Knockout Targets					
R00216[K,p]	Pyruvate (plastid)	DH KO	16,234×	5	<i>GRMZM2G085019</i> , <i>GR-MZM2G085747</i> , <i>GR-MZM2G122479</i> , <i>GR-MZM2G404237</i> , <i>GR-MZM5G886257</i>
R00282[K,c]	Acetyl-CoA carboxylase	KO	28×	1	<i>GRMZM2G000236</i>
Tier 2: Overexpression Targets					
R00883[K,c]	Phosphofructokinase	OE	1.10×	2	<i>GRMZM2G119300</i> , <i>GR-MZM2G149265</i>
R00512[K,c]	Pyruvate kinase	OE	1.10×	4	<i>GRMZM2G067453</i> , <i>GR-MZM2G079944</i> , <i>GR-MZM2G149281</i> , <i>GR-MZM5G801436</i>
R01015[K,c]	Glycolysis (GAPDH)	OE	1.10×	15	<i>GRMZM2G002807</i> , <i>GR-MZM2G017257</i> , +13 isozymes
R00667[K,m]	TCA cycle (mito.)	OE	1.10×	2	<i>GRMZM2G080828</i> , <i>GR-MZM2G119583</i>
R01280[K,p]	Chorismate synthase	OE	1.10×	13	<i>GRMZM2G002614</i> , <i>GR-MZM2G176546</i> , +11 isozymes
Tier 3: <i>o</i>2-Bound Mimicry Targets					
R00342[K,p]	Transketolase (plastid)	MIM	1.10×	11	<i>GRMZM2G035767</i> , <i>GR-MZM2G068455</i> , +9 isozymes
MR00422[K,p]	Lipid metab. (plastid)	MIM	1.10×	15	<i>GRMZM2G002656</i> , <i>GR-MZM2G006620</i> , +13 isozymes
MR00732[K,m]	Biotin metab. (mito.)	MIM	1.10×	4	<i>GRMZM2G377341</i> , <i>GR-MZM5G858094</i> , +2 isozymes
Reference: Lysine Biosynthetic Pathway					
R02292[K,p]	DHDPS	OE	–	1	<i>GRMZM2G027835</i>
R00451[K,p]	DAP decarboxylase	–	–	1	<i>GRMZM2G020446</i>

13 Results 1.13: Flux Variability Analysis

Table S21. FVA maximum achievable flux for key lysine pathway reactions at each developmental stage. All values in $\text{mmol g DW}^{-1} \text{ h}^{-1}$ at 100% optimal biomass.

DAP	R00451[K,p]		cmTransport_C00047		R00715[K,c]	
	WT max	O2 max	WT max	O2 max	WT max	O2 max
6	0.111	0.108	0.037	0.036	0.028	0.024
8	0.000	0.121	0.035	0.040	0.057	0.037
10	0.118	0.000	0.039	0.039	0.051	0.040
12	0.142	0.000	0.047	0.044	0.062	0.059
15	1.336	0.250	0.445	0.083	0.179	0.075
18	5.202	0.000	1.734	0.000	0.398	0.017
22	6.237	5.698	2.079	1.899	0.201	0.188
30	5.368	5.610	1.789	1.870	0.235	0.222

Table S22. Metabolic flexibility per developmental stage: reactions with nonzero max flux, and newly constrained vs newly flexible reactions in O2 relative to WT.

DAP	WT flexible	O2 flexible	Newly constrained in O2	Newly flexible in O2
6	398	919	10	701
8	893	346	719	9
10	934	375	739	18
12	929	935	61	106
15	999	734	409	24
18	976	981	148	46
22	976	979	34	73
30	935	921	0	1